Exception Lab Exercise

1. Write a Java program to accept two integers and perform division. Handle the ArithmeticException when the denominator is zero.

2. Create an integer array and accept an index from the user. Display the element at that index. Handle ArrayIndexOutOfBoundsException.

3. Write a program that accepts two integers and an array index. Perform division and array access, handling the possible exceptions using separate catch blocks.

4. Accept marks for several subjects. If the marks are outside the range 0–100, throw an IllegalArgumentException. Display an appropriate error message.

5. Create a method that can throw two different checked exceptions. Declare both using the throws keyword and handle them in the calling method.

6. Create a package named mypackage containing a class Calculator with methods:

    - add()
    - subtract()
    - multiply()
    - divide()

    Create another class outside the package and import Calculator to perform the operations.

7. Create a package called student containing classes:

    Student – stores student details
    Marks – stores marks
    Result – calculates total and average

    Write a separate Main class outside the package to access these classes and display the result.

8. Create a package shapes containing an interface Shape with a method calculateArea(). Create classes Circle, Rectangle, Triangle that implement the interface. Import the package into another Java program and calculate the areas.

9.